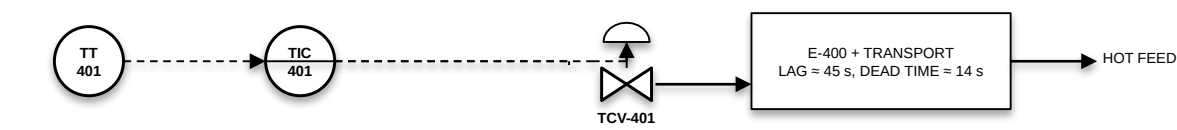
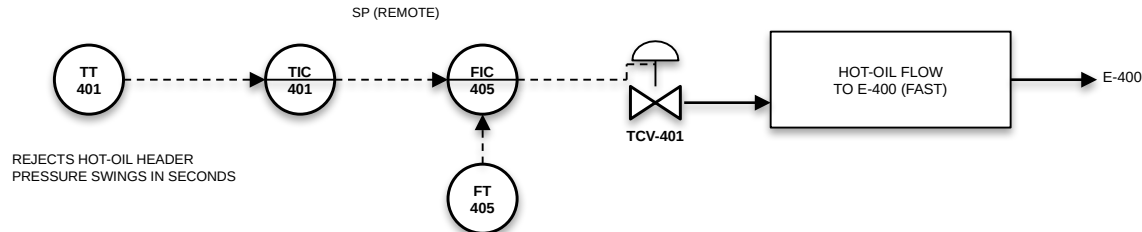


A. CONTROL LOOP CONFIGURATION (SIGNAL FLOW, NOT TO PIPING SCALE)

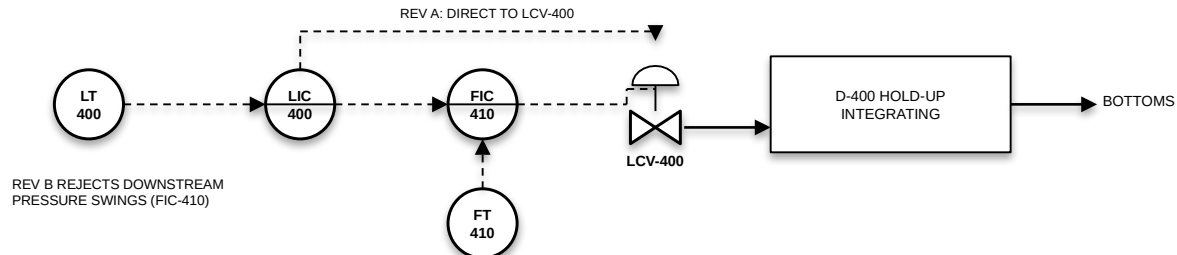
A1 HOT FEED TEMPERATURE — REV A (SINGLE LOOP)



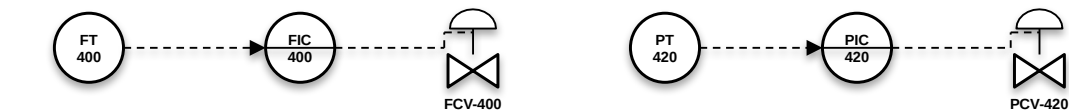
A2 HOT FEED TEMPERATURE — REV B (CASCADE: TIC-401 MASTER, FIC-405 SLAVE)



A3 D-400 LEVEL — REV A: LIC-400 → LCV-400; REV B: LIC-400 → FIC-410 → LCV-400



A4 FEED FLOW FIC-400 → FCV-400 (FC) AND DRUM PRESSURE PIC-420 → PCV-420 (FO)



PIC-420 IS DIRECT ACTING (PRESSURE UP → PCV-420 OPENS). LIC-400 IS DIRECT ACTING (LEVEL UP → LCV-400 OPENS). OTHERS REVERSE.

B. CONTROLLER DATA (DCS), DEFAULT TUNING AND IDENTIFIED PROCESS MODELS

LOOP	SERVICE	RANGE	ACTION	FINAL ELEMENT	ALARM L / H	Kc	Ti s	Td s	BUMP-TEST MODEL (AT DESIGN)
FIC-400	FEED FLOW (COLD SIDE)	0–25 m³/h	REV	FCV-400 FC	5 / 18	0.35	3	0	K 1.8 %/%, τ 2 s, θ 0.6 s
TIC-401	HOT FEED TEMPERATURE	0–150 °C	REV	TCV-401 FC (A)	69 / 79	1.4	55	0	K 0.44–0.65 %/%, τ 45 s, θ 14 s
FIC-405	HOT OIL TO E-400 (REV B)	0–40 t/h	REV	TCV-401 (B)	—	0.4	2.5	0	K = 0.9 %/%, τ 3 s
LIC-400	D-400 LEVEL	0–100 %	DIR	LCV-400 FC (A)	20 / 80	1.5	1200	0	INTEGRATING, 0.002 %/s PER %
FIC-410	BOTTOMS FLOW (REV B)	0–25 m³/h	REV	LCV-400 (B)	8 / 18	0.4	2.5	0	K = 1.5 %/%, τ 2 s
PIC-420	D-400 PRESSURE	0–300 kPa g	DIR	PCV-420 FO	70 / 130	4	180	0	K 0.6 %/%; FAST 10 s + SLOW 250 s

- TUNING NOTES**
- PID ALGORITHM: ERROR IN % OF SPAN, Kc DIMENSIONLESS (%/%), Ti AND Td IN SECONDS, DERIVATIVE ON PV (FILTERED), BUMPLESS, ANTI-WINDUP.
 - DEFAULTS: FLOW LOOPS λ = 3 s; TIC-401 IMC WITH λ = τ; LIC-400 AVERAGING (ARREST TIME = 10 min); PIC-420 λ = 60 s.
 - TIC-401 PROCESS GAIN FALLS AS TCV-401 OPENS (E-400 SATURATES); RE-CHECK TUNING AFTER ANY LARGE CHANGE IN FEED RATE.
 - PIC-420 PRESSURE IS BUFFERED BY THE BOILING HOLD-UP: A FAST SMALL RESPONSE THEN A SLOW LARGE ONE (TWO TIME SCALES).
 - IN REV B TUNE THE SLAVE FIRST (FIC-405 / FIC-410), THEN THE MASTER WITH THE SLAVE IN CAS.

C. SIS CAUSE AND EFFECT (I-1 / I-2)

TAG	CAUSE	SET	DELAY	XV-408 FEED	XV-409 HOT OIL	XV-418 BOTTOMS	RESET
LSHH-416	D-400 LEVEL HIGH-HIGH	90 %	5 s	X	X		LEVEL < 85 %
PSHH-426	D-400 PRESSURE HIGH-HIGH	250 kPa g	3 s	X	X		P < 230 kPa g
LSLL-417	D-400 LEVEL LOW-LOW	5 %	5 s			X	LEVEL > 10 %
HS-ESD	EMERGENCY SHUTDOWN	—	0	X	X	X	MANUAL

SIS NOTES

X = CLOSE (DE-ENERGISE). ALL TRIPS LATCH; XVs ARE RE-OPENED BY THE OPERATOR AFTER RESET (NO AUTOMATIC RESTART).

LSHH-416, LSLL-417, PSHH-426 USE THEIR OWN SENSORS: A FROZEN LT-400 OR BIASED PT-420 DOES NOT DEFEAT THE SIS.

COLD START: I-2 IS LATCHED (LEVEL 0 %) — OPEN XV-418 ONLY AFTER THE LEVEL IS ABOVE 10 %.

PSV-400 (450 kPa g) IS THE LAST LAYER FOR BLOCKED OUTLET / FIRE; RELIEF STUDY HOLD (DWG 001 NOTE 3).

D. FINAL ELEMENTS AND FAIL-SAFE ACTION

TAG	SERVICE	TYPE	FAIL	WHY THIS FAIL POSITION
FCV-400	FEED LET-DOWN	GLOBE, =%	FC	STOPS FEED TO THE DRUM
TCV-401	HOT OIL TO E-400	3-WAY	FC*	*ALL HOT OIL TO BYPASS: NO HEAT
LCV-400	BOTTOMS	GLOBE, =%	FC	KEEPS A LIQUID SEAL (NO GAS BLOW-BY)
PCV-420	DRUM VAPOUR	GLOBE, =%	FO	RELIEVES DRUM TO THE VAPOUR HEADER
XV-408/409/418	SIS SHUT-OFF	BALL	FC	DE-ENERGISE TO TRIP
PCV-430	N2 BLANKET	SELF-ACTING	—	HOLDS ≥ 30 kPa g ON LOSS OF HEAT

E. OPERATING WINDOW AND PRODUCT SPECIFICATIONS

VARIABLE	DESIGN	LOW	HIGH	NOTE
FIC-400 FEED	16.0 m³/h	5	18	SUPPLY 600 kPa g; FCV-400 73 % OPEN AT DESIGN
TIC-401 HOT FEED	74 °C	69	79	TCV-401 30 % TO E-400 AT DESIGN
PIC-420 DRUM	102.7 kPa g	70	130	PSHH 250; PSV 450; N2 30
LIC-400 DRUM	50 %	20	80	LSHH 90; LSLL 5
AI-410 BOTTOMS C4	26.5 wt%	—	28	BOTTOMS SPEC ≤ 28 wt% C4
AI-420 VAPOUR C4	76.8 wt%	75	—	VAPOUR SPEC ≥ 75 wt% C4
AI-400 N2 IN VAPOUR	0 mol%	—	5	START-UP PURGE COMPLETE < 1 %

- GENERAL NOTES**
- THIS SHEET IS THE CONTROL NARRATIVE FOR DWG FD-400-PID-001. TAGS, LOOPS AND SETTINGS MATCH THE OTS MODEL (FLASH DRUM VER 1).
 - MODEL IDENTIFICATION BY OPEN-LOOP STEP OF THE CONTROLLER OUTPUT AT DESIGN (NOISE OFF). VALUES VARY WITH THE OPERATING POINT.
 - PROCESS GAINS K ARE IN % OF TRANSMITTER SPAN PER % OF CONTROLLER OUTPUT.
 - TUNING, ALARM AND TRIP SETTINGS ARE TRAINING VALUES — NOT FOR A REAL PLANT WITHOUT HAZOP / LOPA REVIEW.

REV	DESCRIPTION	DATE	
0	ISSUED FOR TRAINING (OTS FLASH DRUM VER 1)	24-SEP-26	
CHEM. ENG., CHULALONGKORN UNIVERSITY — OTS			
D-400 CONTROL, LOOP DATA & SIS C&E			
DRAWN: A. SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 24-SEP-26
DWG NO. FD-400-PID-002	REV 0	SCALE N.T.S.	SHEET 2 OF 2